

17. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps $u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ into $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and maps $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ into $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$. Use the fact that T is linear to find the images under T of $3u$, $2v$, and $3u+2v$.

Solution: From defn. of linear transformation, for all x and y in $\mathbb{R}^2$,
 1. $T(x+y) = T(x) + T(y)$ 2. $T(cx) = cT(x)$, for any arbitrary scalar c .
 Given $T(u) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $T(v) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. $T(3u) = 3T(u)$ (From condn. 2)
 $= 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ (Scalar multiplication)

$$T(2v) = 2T(v) \text{ (From condn. 2)} = 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$T(3u+2v) = T(3u) + T(2v) \text{ (By condn. 1)} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix} \text{ (Vector addn.)}$$

19. Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $y_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, and $y_2 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$, and let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps e_1 into y_1 and maps e_2 into y_2 . Find the images of $\begin{bmatrix} 5 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

Solution: Given, $T(e_1) = y_1$ and $T(e_2) = y_2$. We shall use the two conditions from the defn. of a linear transformation.

$$\begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 5+0 \\ 0+(-3) \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \end{bmatrix} \text{ (Vector addition)} = \begin{bmatrix} 5 \cdot 1 \\ 5 \cdot 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

$$+ \begin{bmatrix} (-3) \cdot 0 \\ (-3) \cdot 1 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-3) \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$= 5e_1 + (-3)e_2$$

$$\therefore T \begin{bmatrix} 5 \\ -3 \end{bmatrix} = T(5e_1 + (-3)e_2) = T(5e_1) + T((-3)e_2) \text{ (Condition (i))}$$

$$= 5T(e_1) + (-3)T(e_2) \text{ (Condition (ii))}$$

$$= 5y_1 + (-3)y_2 = 5 \begin{bmatrix} 2 \\ 5 \end{bmatrix} + (-3) \begin{bmatrix} -1 \\ 6 \end{bmatrix} = \begin{bmatrix} 10 \\ 25 \end{bmatrix} + \begin{bmatrix} 3 \\ -18 \end{bmatrix} \text{ (Scalar mult.)}$$

$$= \begin{bmatrix} 13 \\ 7 \end{bmatrix} \text{ (Vector addition)}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1+0 \\ 0+x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \end{bmatrix} \text{ (Vector addition)}$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$= x_1 e_1 + x_2 e_2$$